

Computer Networks Course Project

University Campus Network Design

Description: Design a network for a small university campus.

- The university has several faculties, at minimum four, and one Scientific Computation Center.
- Scientific Computation Center contains several servers that all connected with a switch.
- Each faculty is composed of several floors, at a minimum of four.
- Each floor is composed of several labs, at minimum five.
- Each lab contains at least ten PCs, all these PCs are connected with switch.
- All the switches of the labs on the same floor are connected with a switch.
- All the floors are connected by a switch that connects with the main router of the university.
- The university router connects all the faculties and the Scientific Computation Center with each other and with the world wide web.

Please, Apply the following points:

- Use Classless to distribute the public IP addresses to all sub-networks.
- Use the mesh and star topology in the labs design.
- Write three test scenarios to evaluate your network in addition to these scenario example:
- Scenario description: Pc number 0 on the first floor in the second lab in the faculty of computers & Artificial Intelligence want to access data on the first server in the Scientific Computation Center.
- Source IP address: Pc 0 IP address.
- Destination IP address: Server 0 IP address.
- Apply VLAN configuration in your network and examine it in a separate scenario.

Bonus: RIP Routing Protocol (3 grades)

RIP (Routing Information Protocol) is implemented for routing within the university network.

Used to route data between faculties and the Scientific Computation Center.

Project supervised by:

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